

HW 1

(1a) Over here we have a perceptron that has been given to us:

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x})$$

and we also have:

$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$

Now, when we multiply $\mathbf{w}^T \mathbf{x}$, we get the following:

$$\mathbf{w}^T \mathbf{x} = w_0 + w_1 x_1 + w_2 x_2$$

Now, the perceptron will predict its value, after analyzing whether this value is positive or negative, so if we have:

$$w_0 + w_1 x_1 + w_2 x_2 > 0$$

So, we can say that the prediction is +1.

However, if we have this:

$$w_0 + w_1 x_1 + w_2 x_2 < 0$$

Then, we can say that the prediction is -1.

Hence, we can say that the exact place where the prediction changes is:

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

(which is also called the decision boundary)

Now, since we want to show this as a boundary, we need to rearrange this equation in the following form:

$$x_2 = ax_1 + b \tag{1}$$

and we have:

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

$$w_2 x_2 = -w_0 - w_1 x_1$$

$$x_2 = \frac{-w_0 - w_1 x_1}{w_2} \tag{2}$$

By looking at equations 1 and 2 we can say that $a = -\frac{w_1}{w_2}$, $b = -\frac{w_0}{w_2}$.

(1b) First case: $\mathbf{w} = [1, 2, 3]^T$

From part a, we can say that our decision boundary is where $\mathbf{w}^T \mathbf{x} = 0$, so let's plug in $w_0 = 1$, $w_1 = 2$ and $w_2 = 3$, so we must get $1 + 2x_1 + 3x_2 = 0$, so for $3x_2 = -2x_1 - 1$,

$$x_2 = -\frac{2}{3}x_1 - \frac{1}{3},$$

which is basically the line that we need to draw and to make it easier, we need to find 2 points on it.

Now, if $x_1 = 0$, we get $x_2 = -\frac{1}{3}$ and if $x_2 = 0$, we get $x_1 = -\frac{1}{2}$, so our 2 points are $(0, -\frac{1}{3})$ and $(-\frac{1}{2}, 0)$, now we need to connect the points to make the graph.

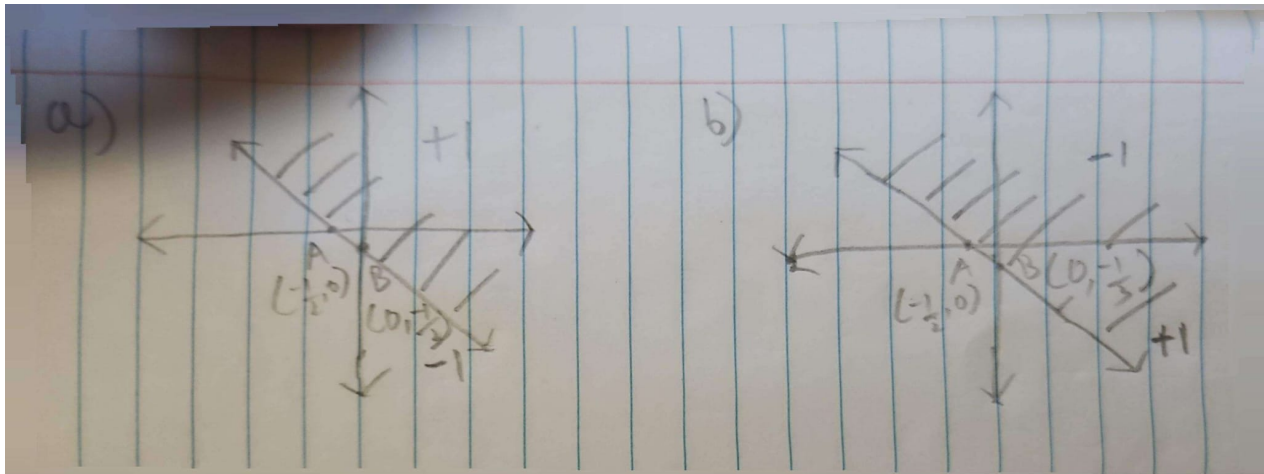
Now, we can say that our perceptron is $h(\mathbf{x}) = \text{sign}(1 + 2x_1 + 3x_2)$, so let's choose a point that does not lie on the line, for example we have the point $(0, 0)$, so let us plug that point in the equation.

$$1 + 2(0) + 3(0) = 1$$

Since $1 > 0$, we can say that whichever side contains the point $(0, 0)$ is the positive region and since the boundary crosses the x_2 axis at $-\frac{1}{3}$, we can say that the origin is above the line. Thus, above the line is $+1$ and below the line is -1 .

Second case: $\mathbf{w} = -[1, 2, 3]^T$

Now, we have $\mathbf{w} = [-1, -2, -3]^T$, the boundary becomes $-1 - 2x_1 - 3x_2 = 0$, which means that $1 + 2x_1 + 3x_2 = 0$, which means that it is the same line as part 1, however, if we test the origin, we get $-1 - 2(0) - 3(0) = -1$, which is less than 0, so we can say that the regions have reversed. Thus, for this part, above the line is -1 and below the line is $+1$.



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(2a) Let us first define a variable, let's say:

$$z = -y_n \mathbf{w}^T \mathbf{x}_n$$

Which makes our expression $\ln(1 + e^z)$, which means that if we get differentiated this with respect to z , then we get:

$$\frac{e^z}{1 + e^z}$$

Now, we need to differentiate z with respect to $\mathbf{w}$, and we already know that the derivative of $\mathbf{w}^T \mathbf{x}_n = \mathbf{x}_n$, so it means that the derivative of $-y_n \mathbf{w}^T \mathbf{x}_n$ is $-y_n \mathbf{x}_n$.

Next, let's multiply the 2 derivatives:

$$\left(\frac{e^z}{1 + e^z} \right) (-y_n \mathbf{x}_n)$$

So, now let us substitute $z = -y_n \mathbf{w}^T \mathbf{x}_n$, so we get:

$$\left(\frac{e^{-y_n \mathbf{w}^T \mathbf{x}_n}}{1 + e^{-y_n \mathbf{w}^T \mathbf{x}_n}} \right) (-y_n \mathbf{x}_n)$$

Now, let's multiply the numerator and denominator by $e^{y_n \mathbf{w}^T \mathbf{x}_n}$, which will give us

$$\frac{1}{1 + e^{y_n \mathbf{w}^T \mathbf{x}_n}},$$

which means that derivative of 1 training example will be

$$-\frac{y_n \mathbf{x}_n}{1 + e^{y_n \mathbf{w}^T \mathbf{x}_n}}.$$

Now, as we can see that our original loss averages over all N training examples:

$$E(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N \ln \left(1 + e^{-y_n \mathbf{w}^T \mathbf{x}_n} \right)$$

So, we will just take the derivative of all the terms over here, which leaves us with the following equation:

$$\nabla E(\mathbf{w}) = -\frac{1}{N} \sum_{n=1}^N \frac{y_n \mathbf{x}_n}{1 + e^{y_n \mathbf{w}^T \mathbf{x}_n}}$$

(2b) Over here, we know that the decision boundary occurs when the predicted probability is basically equal to the classification threshold, which is actually 0.5 in this case.

So, now by using the sigmoid function we get the following:

$$\frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}} = 0.5$$

Which gives us:

$$1 + e^{-\mathbf{w}^T \mathbf{x}} = 2$$

$$e^{-\mathbf{w}^T \mathbf{x}} = 1$$

Which means that $\mathbf{w}^T \mathbf{x} = 0$

Hence, we can say that the decision boundary is determined by $\mathbf{w}^T \mathbf{x} = 0$, which means that it is a linear equation in $\mathbf{x}$, so the decision boundary is linear, and even though the sigmoid function

is non-linear, it transforms the linear value $\mathbf{w}^T \mathbf{x}$ into a value that's between 0 and 1, and that's why it does not change the fact that the classification boundary itself is determined by the linear expression $\mathbf{w}^T \mathbf{x}$.

- (2c) Over here we can see that for the new threshold, the new decision boundary occurs when $\theta(\mathbf{w}^T \mathbf{x}) = 0.9$, so by using the sigmoid function we get:

$$\frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}} = 0.9$$

$$1 + e^{-\mathbf{w}^T \mathbf{x}} = \frac{1}{0.9}$$

$$1 + e^{-\mathbf{w}^T \mathbf{x}} = \frac{10}{9}$$

$$e^{-\mathbf{w}^T \mathbf{x}} = \frac{1}{9}$$

Now, if we take log on both the sides then we get the following:

$$-\mathbf{w}^T \mathbf{x} = \ln\left(\frac{1}{9}\right)$$

which leaves us with:

$$\mathbf{w}^T \mathbf{x} = \ln(9)$$

Hence, we can say that the new decision boundary is determined by $\mathbf{w}^T \mathbf{x} = \ln(9)$, which is a linear equation in $\mathbf{x}$ and changing the classification threshold only shifts the location of the decision boundary over here, which means that the boundary still remains linear.

- (2d) In my opinion, the essential property of logistic regression that results in the linear decision boundary is that the sigmoid function is monotonic, which means that as the value of $\mathbf{w}^T \mathbf{x}$ increases, the output of the sigmoid function also increases and because of that any fixed classification threshold corresponds to a fixed value of $\mathbf{w}^T \mathbf{x}$, which means that the decision boundary can be written in the form of $\mathbf{w}^T \mathbf{x} = \text{constant}$.

Now, since $\mathbf{w}^T \mathbf{x}$ is linear in $\mathbf{x}$, we can say that the decision boundary will remain linear, as the sigmoid function changes the output into a probability, but it does not change the linear form of the boundary.

- (3) Over here we have

$$X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{m \times n},$$

where each $x_i \in \mathbb{R}^m$ and

$$Y = \begin{bmatrix} y_1^T \\ y_2^T \\ \vdots \\ y_n^T \end{bmatrix} \in \mathbb{R}^{n \times p},$$

where each $y_i \in \mathbb{R}^p$ and we actually need to show that

$$XY = \sum_{i=1}^n x_i y_i^T.$$

First, the matrix X is built using the vectors as x_i as columns and matrix Y is built using the transposes y_i^T as rows, so that's why X and Y have sizes $m \times n$ and $n \times p$, and hence XY has the size as $m \times p$ [which is a result of $(m \times n)$ and $(n \times p)$].

Now, if we look at one of the terms, which is $x_i y_i^T$ on the right side, we know that $x_i \in \mathbb{R}^m$, we can say that x_i is an $m \times 1$ column vector and $y_i \in \mathbb{R}^p$, then y_i^T is basically a $1 \times p$ row vector, hence we say that $x_i y_i^T$ has size $m \times p$, that means every term in $\sum_{i=1}^n x_i y_i^T$ is also an $m \times p$ matrix, and that's why the dimensions make sense.

Now, let's just try to understand why the equality holds true, so we write X and Y next to each other and we get:

$$XY = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1^T \\ y_2^T \\ \vdots \\ y_n^T \end{bmatrix}$$

When we multiply this block by block, we basically get:

$$XY = x_1 y_1^T + x_2 y_2^T + \cdots + x_n y_n^T$$

Which we can write in the following way:

$$XY = \sum_{i=1}^n x_i y_i^T.$$

- (4) Over here, we can say that, in the binary case, softmax has only 2 classes, so let us assume that the 2 classes scores are y_1 and y_2 .

For class 1, the softmax function is:

$$\text{softmax}(y_1) = \frac{e^{y_1}}{e^{y_2} + e^{y_1}}$$

Now, if we divide both the numerator and the denominator by e^{y_1} , we basically get:

$$\text{softmax}(y_1) = \frac{1}{1 + e^{y_2 - y_1}}$$

This can be basically rewritten as:

$$\text{softmax}(y_1) = \frac{1}{1 + e^{-(y_1 - y_2)}}$$

Now, if we have $z = y_1 - y_2$, then we get:

$$\text{softmax}(y_1) = \frac{1}{1 + e^{-z}}$$

(which is exactly the sigmoid function)

Hence, we can say that in the binary case, the softmax function is equivalent to the sigmoid function and that the sigmoid function can be viewed as the binary version of softmax, where the input is actually the difference between 2 class scores.